

BUSINESS PROBLEM STATEMENT

A digital marketing company ran an advertising experiment across 588,101 users to evaluate the effectiveness of their ad campaign. Users were split into two groups — one group was shown advertisements while the other was shown a Public Service Announcement (PSA) as a control. With an overall conversion rate of just 2.52%, the marketing team needs a data-driven answer to whether the ads are actually driving conversions or whether the results could have occurred by chance. The team also wants to understand which days and hours drive the highest conversions so they can optimize their ad spend.

Core Business Question

“Are the ads statistically proven to increase conversions compared to the PSA group — and when should the company run ads to maximize conversion rates?”

Deliverables

- **Data Preparation & Exploration (Python):** Load and clean the raw dataset, drop unused columns, verify data quality (nulls, duplicates), and explore group sizes, conversion counts, and behavioral patterns across days and hours.
- **Statistical Analysis (Python):** Conduct a two-proportion z-test to determine whether the difference in conversion rates between the ad and PSA groups is statistically significant. Calculate the z-score, the 95% confidence interval, and the lift to support the business recommendation.
- **Visualization & Insights (Power BI):** Build an interactive dashboard showing conversion rate comparison, day-of-week performance, and hourly trends to help the marketing team decide when and how to run future campaigns.
- **Report and Recommendation:** Summarize the statistical findings and provide an actionable business recommendation on whether to continue, scale, or revise the ad campaign, including the best day and hours to run it.